DAVID MAYOWA ONAIYEKAN



AI/ML Engineer

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Portfolio: devo002.github.io | GitHub: github.com/devo002

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PROFESSIONAL SUMMARY

AI / Machine Learning Engineer with an M.Sc. in Artificial Intelligence, combining hands-on research in Deep Learning, Computer Vision, and Generative AI with production engineering experience building end-to-end LLM applications, RAG systems, and agentic AI workflows. Skilled in training and evaluating models (PyTorch, Vision Transformers, CNNs) as well as designing and deploying production-ready AI systems with LangChain, LangGraph, LlamaIndex, FastAPI, and vector databases. Experienced integrating AI into real-world business processes and deploying solutions with Docker and AWS.

TECHNICAL SKILLS

Languages: Python (Advanced), JavaScript, TypeScript.

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, Numpy, Pandas.

Computer Vision: OpenCV, Image Processing, CNNs, Vision Transformers, OCR, GANs.

LLMs & AI Engineering: OpenAI, Anthropic, Llama, Hugging Face, LangChain, LangGraph, LlamaIndex, RAG, Prompt Engineering, MCP.

Databases & Vector Databases: PostgreSQL, MySQL, MongoDB, Qdrant, Pinecone, ChromaDB.

MLOps, Cloud & Deployment: AWS, Docker, FastAPI, CI/CD, GitHub Actions.

Automation & Workflows: n8n, Make, AI Workflow Orchestration.

AI ENGINEERING PROJECTS

Smart Dispatcher — Agentic Ticket Triage & Dispatch | Live Demo • GitHub 04/26 - 07/26

- Developed and deployed a production-ready agentic AI system that automatically triages support tickets and generates primary and fallback dispatch plans, reducing manual decision-making while keeping humans in the approval loop.
- Designed a 6-node LangGraph workflow with a self-correction loop that detects failed repair plans and automatically retrieves firmware-based workarounds via RAG, without additional LLM prompting.
- Engineered guardrails into the RAG retrieval pipeline — metadata-filtered vector search and regex-based extraction guards — to eliminate embedding confusion between similar error codes and false-positive part requirements, validated with a 17-case evaluation suite.
- Built a React/FastAPI dashboard with real-time SSE streaming and full LangSmith observability (per-step latency, token usage, cost tracking) to test, monitor, and evaluate agent performance over time.
- Connected the inventory system to a live Google Sheet via the Google Sheets API, so any stock update made directly in the sheet is automatically reflected in the agent's decisions.

Tech: Python, LangGraph, Claude (Anthropic SDK), ChromaDB, FastAPI, LangSmith, Google Sheets API, Railway, Tailwind, Pydantic.

RAG-Powered Document Chat System | Live Demo • GitHub 02/26 - 05/26

- Developed a production-deployed RAG application enabling users to upload PDFs and receive streaming, grounded answers with page-level source citations.
- Implemented a hybrid retrieval pipeline combining dense vector embeddings and BM25 keyword search, fused via Reciprocal Rank Fusion (RRF) and reranked by Claude to improve answer grounding across multiple documents.
- Designed a RAGAS-inspired evaluation framework scoring Faithfulness, Answer Relevancy, and Correctness with all metrics streamed live into the UI.

Tech: Python, FastAPI, LlamaIndex, SSE streaming, RAG, Qdrant, Docker, Railway, GitHub CI/CD.

AI Ticket Bridge Agent — Teams to Jira Automation | Github 02/26 - 04/26

- Developed an AI bridge agent that classifies Microsoft Teams conversations, enriches them with CRM context, deduplicates against Jira, and reduces manual ticket creation to a one-click approval workflow.
- Optimized a multi-stage agent pipeline to process 70+ support messages using fewer than four Claude API calls while streaming progress to reviewers in real time.
- Built a review dashboard enabling human validation of AI recommendations before synchronizing approved actions with Jira and Microsoft Teams.

Tech: Python, Flask, Javascript, Claude, Jira REST API, Docker, Git.

Full-Stack Budget & Expense Tracker | Live Demo • Github 01/26 - 03/26

- Developed and deployed a full-stack personal finance tracker that allows users to record transactions, monitor weekly spending limits, and calculate monthly savings dynamically.
- Implemented a RESTful backend using Next.js API routes and Prisma ORM to manage transaction creation, deletion, and budget data stored in a PostgreSQL database.
- Built a responsive dashboard with real-time financial summaries and deployed using Vercel with a Neon cloud database for cross-device access.

Tech: TypeScript, Next.js, Prisma Studio, PostgreSQL (Neon), Git, Vercel.

RESEARCH EXPERIENCE

Machine Learning Researcher
Pattern Recognition Lab, FAU | Github 06/25 – 01/26

- Fine-tuned and evaluated alternative backbone architectures for the AFFGANwriting style encoder to improve capture of writer-specific handwriting features, benchmarking against the original baseline architecture.
- Extended the system beyond word-level synthesis to line-level handwriting generation, integrating and adapting an existing line-generation architecture to support full-sentence output.
- Deployed the handwriting generation pipeline as a web application, enabling users to select a writing style, input custom text, and instantly generate a corresponding handwriting image.

- Integrated TrOCR into a teacher–student distillation framework to evaluate and improve the readability of generated handwriting, improving OCR accuracy by 20% over the baseline.

  Tech: PyTorch, Python, GANs, CNN, ViT, TrOCR, Streamlit, CUDA.

Computer Vision Researcher
Institute for Factory Automation and Production Systems (FAPS) | Live Demo • Github
12/23 – 01/25

- Collaborated within a research team to execute systematic data cleaning on thousands of industrial coil-winding images — manually correcting label noise on a representative subset and leveraging Cleanlab Studio's autofix to propagate corrections across the remaining dataset, establishing a high-fidelity baseline for downstream model training.
- Trained and evaluated fully-supervised baseline classifiers (DINOv2-L and EfficientNetV2s) to establish reference macro-F1 performance, benchmarking against SSL (FixMatch/MixMatch) gains across four label-fraction splits (10%, 25%, 50%, 100%).
- Ran systematic hyperparameter optimization with Optuna across backbone/algorithm combinations, tuning learning rate, batch size, and augmentation parameters, and tracked all trials via a persistent SQLite results database for reproducibility.
- Designed and implemented a config-driven multi-label SSL training pipeline supporting both FixMatch and MixMatch algorithms, with modular backbone integration for DINOv2-L and EfficientNetV2s.
- Achieved a 90% macro-F1 average score with FixMatch, demonstrating SSL's value for reducing annotation cost.

Tech: PyTorch, DINOv2-L, EfficientNetV2-L, FixMatch, Optuna, WandB, Docker.

EDUCATION

Master of Science in Artificial Intelligence 01/26
Friedrich-Alexander-Universität Erlangen-Nürnberg, Erlangen, Germany

Bachelor of Technology in Computer Science 11/18
Federal University of Technology, Minna.

LANGUAGES

- English (Native)
- German (Intermediate)